

Correction and Conditional Reformulation of the Mod-16 Expansion Bound in Collatz Dynamics

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Abstract

We correct a significant arithmetic error in the previously claimed absolute upper bound $\tau(n) \leq C \log n$ for Collatz trajectories. The error arises from counting four-step blocks as two-step blocks, leading to an undercount of the worst-case amplification. We show that the correct four-step net amplification is $(3/2)^3 \times (1/2) = 27/16 > 1$. Consequently, the constant- C absolute bound is invalid. We reformulate P36 as a conditional bound that depends on the choice of the deterministic window length K , and we provide the corrected framework for future analysis of state-machine expansion envelopes.

1. The Original Claim

In earlier versions of this work, the following proposition was stated as rigorously proven:

Original P36 (erroneous): There exists a constant $C = 4 / \log_2(4/3) \approx 9.64$ such that for all n , $\tau(n) \leq C \log_2 n$, where $\tau(n)$ is the number of steps until the trajectory first falls below n .

Proof attempt (flawed): Since the mod-16 deterministic transition matrix guarantees at least one division by 2 in every 4-step block, and the maximum single-step amplification is $3/2$, the net amplification over 4 steps is at most:

$$(3/2) \times (1/2) = 3/4 < 1$$

This is arithmetically incorrect.

2. The Correction

A 4-step block consists of four individual transformations. If the worst case is three odd steps (amplification by approximately $3/2$ each) and one even step (division by 2), the net amplification is:

$$(3/2)^3 \times (1/2) = 27/16 = 1.6875 > 1$$

The original calculation treated the 4-step block as if it had only two steps: one amplification and one division. This is a factor of 4 error in step counting.

3. Implications for the Constant-C Bound

Since $27/16 > 1$, a 4-step block can have net amplification. Therefore:

- There is no immediate guarantee that the trajectory decreases after any fixed number of steps.
- The claimed constant $C = 4/\log_2(4/3)$ is invalid.

- The proposition "there exists an absolute constant C such that $\tau(n) \leq C \log^{\frac{1}{2}} n$ for all n " is not established by this argument.

4. Reformulation as a Conditional Bound (P36 Revised)

Revised P36 (conditional): For any chosen deterministic window length K (i.e., working modulo 2^K), the maximal possible amplification over K steps is bounded by the maximum product of local transition factors along the state machine's longest path with net growth > 1 . Let G_K be the maximum over all K -step paths of the product of $(3 / 2^{v_i})$ where $v_i \geq 1$.

If $G_K < 1$ for some K , then:

$$\tau(n) \leq K \times \log_{1/G_K} n = [K / \log(1/G_K)] \log n$$

Status: Conditional upon determining G_K for a finite K . For $K=4$, $G_4 = 27/16 > 1$, so this K is insufficient.

Future direction: Compute G_K for $K=5, 6, \dots$ (i.e., mod 32, mod 64, etc.) until the first K with $G_K < 1$ appears. If such a K exists, an absolute bound can be recovered. If no such K exists, then no constant- C absolute bound is possible from finite-state arguments alone.

5. Connection to Broader Lessons

This correction is significant for three reasons:

1. **Methodological honesty:** It demonstrates that the framework is self-correcting and falsifiable. A flawed constant was identified, corrected, and the proposition was downgraded rather than hidden.
2. **Boundary awareness:** The $m=1$ boundary case in $\sigma(2m-1) = 2 + \sigma(3m-1)$ was also corrected ($\sigma(1)=0$, not 3). Together, these corrections show that rigorous Collatz analysis requires meticulous attention to the base cases and worst-case path enumeration.
3. **State machine limits:** The correction reveals that the mod-16 state machine alone is insufficient to prove a global boundedness constant. Higher moduli (mod 2^K , $K \geq 5$) must be examined, or entirely different methods must be employed.

6. Conclusion

The corrected P36 no longer claims an absolute constant- C bound. Instead, it provides a conditional framework: if a sufficiently long deterministic window with net contraction $G_K < 1$ exists, then a bound follows; otherwise, finite-state methods are insufficient for global boundedness. This honest reformulation strengthens the overall framework by clarifying its assumptions and boundaries.

References

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